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Details Questions

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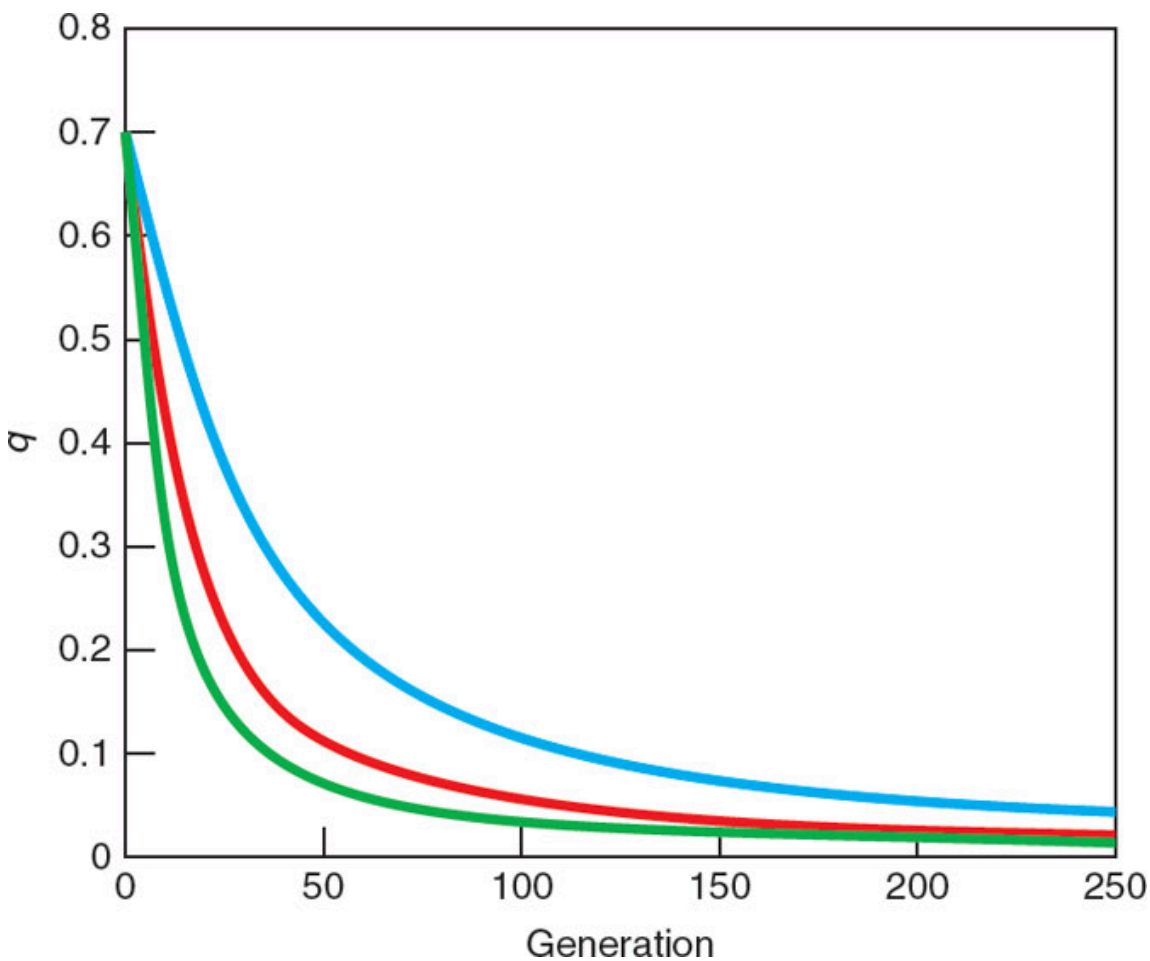
Question 1

Chapter 21, Solved Problems III: Three different genes (red, blue, and green) each have two alleles; one allele for each gene has deleterious recessive effects. The relative fitness of the recessive homozygotes for one of these genes is 0.9, for another it is 0.8, and for the third it is 0.7. The following graph depicts changes in the frequencies of these alleles in populations of infinite size over time; in each case, the frequency of the allele in question (q) is 0.7 at the beginning of the experiment.



Question 2 1.3 pts

For each gene, calculate Δq between the parental generation and the first generation of progeny (q = frequency of the recessive deleterious allele). Assume that the relative fitnesses of heterozygotes and of homozygotes for the dominant allele are 1.0. You do not yet need to know which gene corresponds to which color in the graph.

The formula for Δq is:

$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\bar{W}}$$

where $\bar{W}$ is the mean fitness (the sum of the relative contributions of each genotype to the next generation.

Gene A has the relative fitness of the recessive homozygotes as 0.9. What is the Δq for gene A: [geneA]

Gene B has the relative fitness of the recessive homozygotes as 0.8. What is the Δq for gene B: [geneB]

Gene C has the relative fitness of the recessive homozygotes as 0.7. What is the Δq for gene C: [geneC]

Give your answers rounded to four decimal places.

Show Answers for geneA

-0.0155

Question 0.9 pts

How does genetic drift influence genotypes and phenotypes in small populations, relative to large populations?

Higher frequency of deleterious genetic variation
Allele frequencies consistent with HWE
Higher rates of rare disease
Higher amounts of neutral genetic variation

Question 0.9 pts

Some populations are small and some populations are large enough to be approximated by infinity.

Select the term below that describes a population that has undergone a rapid and drastic reduction in size:

Population reduction
Size squeeze
Genetic bottleneck
Finite population

Question 0.9 pts

Small populations respond efficiently to natural selection pressures

True
False

Question 1 pts

What population genetic process is a small population highly susceptible to (two words)?

Genetic drift

genetic drift

Genetic Drift

geneticdrift

genetic_drift

Question 1 pts

Many years ago, a small population of birds moved from the mainland to an island.

The mainland bird population contained a diverse mix of colors, but the migrated birds were only blue and yellow.

Over time, the yellow allele was lost from the island population due to genetic drift.

Would you expect the migrant population to initially be in Hardy Weinberg Equilibrium?

yes
no
cannot be determined

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